

Chapter 2

Projectiles and Charged Particles

I covered this chapter in 3.5 fifty-minute lectures.

The topics of Chapter 2 are two relatively straightforward applications of Newton's laws of motion — the motion of projectiles subject to air resistance and of a charged particle in a magnetic field. Both of these are interesting topics that your students may not have studied in detail before. In particular, while air resistance is often mentioned in most freshman physics courses, it is usually mentioned only to be neglected. The inclusion of air resistance gives us a chance to discuss the solution of several non-trivial differential equations and (in the case of quadratic resistance) is our first chance to confront differential equations that can only be solved numerically. (My policy with respect to numerical solution of differential equations is to endorse no specific software, but rather to urge the use of whatever software — Mathematica, Maple, Matlab — is available. Some of the simpler problems can be done with a spreadsheet such as Excel, as can many of the harder problems if you can help your students with the numerical methods needed.) The motion of a charged particle in a magnetic field is another interesting problem and it gives me the opportunity to introduce the use of complex numbers in solving real problems.

These two topics are a natural context to introduce several other mathematical techniques besides differential equations and complex numbers. For example, in Section 2.3 Taylor's series emerges as a natural way to account for weak air resistance and Section 2.4 introduces the hyperbolic functions $\sinh$, $\cosh$, and $\tanh$ (of which most of my students claimed complete ignorance). As usual, the end-of-chapter problems offer several opportunities for the students to get up to speed on any math that they need help with. For instance, Problems 2.33 and 2.34 provide a crash course in the hyperbolic functions, and Problems 2.45 through 2.51 should refresh memories about complex numbers. There are several problems requiring the use of a computer to find orbits and ranges of projectiles subject to air resistance.

Solutions to Problems for Chapter 2

2.1 ** According to (2.7), $f_{\text{quad}}/f_{\text{lin}} = vD\gamma/\beta$, so this equals 1 when $v = 1/(D\gamma/\beta)$, which is roughly 1 cm/s if $D = 7$ cm. Since baseballs generally travel much faster than 1 cm/s, it

is usually a very good approximation to ignore f_{lin} . For a beachball with $D = 70$ cm, the corresponding speed is about 1 mm/s.

2.2 * According to Stokes's law $f_{\text{lin}} = (3\pi\eta D)v$, which has precisely the form $f_{\text{lin}} = bv$ if we define $b = \beta D$ and $\beta = 3\pi\eta = 3\pi(1.7 \times 10^{-5} \text{ N}\cdot\text{s}/\text{m}^2) = 1.6 \times 10^{-4} \text{ N}\cdot\text{s}/\text{m}^2$.

2.3 * (a) From (2.84) and (2.82), $f_{\text{quad}}/f_{\text{lin}} = (\kappa\rho Av^2)/(3\pi\eta Dv)$. With $\kappa = 1/4$ and $A = \pi D^2/4$, this becomes $\rho Dv/(48\eta)$ or $R/48$, with R given by (2.83).

(b) With the given numbers, $R = 1.1 \times 10^{-2}$ and it is very safe to neglect the quadratic drag.

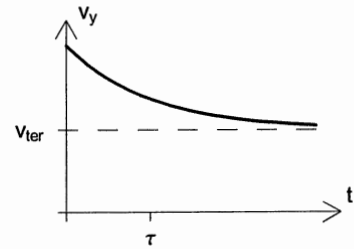
2.4 ** (a) In a short time dt the projectile moves a distance vdt , and the front sweeps out a cylinder of volume $Avdt$. Therefore the mass of fluid encountered is $\rho Avdt$, and the rate at which mass is swept up is ρAv .

(b) If a mass $\rho Avdt$ is accelerated from 0 to v in time dt , the rate of change of its momentum is ρAv^2 . This is, therefore, the forward force on the fluid and, hence, the backward force on the projectile.

(c) Since $A \propto D^2$, it follows that $f_{\text{quad}} = \kappa\rho Av^2 = cv^2$, where $c = \kappa\rho A \propto D^2$. For a sphere in air, $\kappa = 1/4$, $A = \pi D^2/4$, and $\rho = 1.29 \text{ kg}/\text{m}^3$, so $f_{\text{quad}} = (\kappa\rho\pi D^2/4)v^2 = cv^2$, where $c = \gamma D^2$ and

$$\gamma = \kappa\rho\pi/4 = \frac{1}{4} \times (1.29 \text{ kg}/\text{m}^3) \times \pi/4 = 0.25 \text{ N} \cdot \text{s}^2/\text{m}^4.$$

2.5 * With $v_y > v_{\text{ter}}$, the drag force is greater than the weight, and the net force is upward. Thus the projectile slows down, with v_y approaching v_{ter} as $t \rightarrow \infty$. This is clear from Eq.(2.30), as shown in the plot.



2.6 * (a) If we insert the Taylor series for $e^{-t/\tau}$ into (2.33), we get

$$v_y(t) = v_{\text{ter}} [1 - e^{-t/\tau}] = v_{\text{ter}} \left[1 - \left(1 - \frac{t}{\tau} + \frac{t^2}{2\tau^2} - \cdots \right) \right].$$

The first two terms on the right cancel, and, if t is sufficiently small, we can neglect terms in t^2 and higher. This leaves us with

$$v_y(t) \approx v_{\text{ter}} t/\tau = gt$$

where to get the second equality I replaced v_{ter} by $g\tau$ as in (2.34).

(b) Putting $v_{y0} = 0$ into (2.35) and then inserting the Taylor series for the exponential, we find:

$$y(t) = v_{\text{ter}}t - v_{\text{ter}}\tau [1 - e^{-t/\tau}] = v_{\text{ter}}t - v_{\text{ter}}\tau \left[1 - \left(1 - \frac{t}{\tau} + \frac{t^2}{2\tau^2} - \cdots \right) \right].$$

On the right side, the second and third terms cancel, as do the first and fourth. If we neglect all terms beyond t^2 , this leaves us with $y(t) \approx v_{\text{ter}}t^2/(2\tau) = \frac{1}{2}gt^2$, since $v_{\text{ter}} = g\tau$.

2.7 * With $F = F(v)$, Newton's second law, $F = m dv/dt$, can be rewritten as $dt = m dv/F(v)$, which can be integrated to give the advertised result. If $F = F_0$, a constant, the integral gives $t = (m/F_0)(v - v_0)$ or $v = v_0 + at$, where $a = F_0/m$. This is the well known kinematic formula for v when a is constant.

$$\mathbf{2.8 \star} \quad t = m \int_{v_0}^v \frac{dv'}{-cv'^{3/2}} = \frac{2m}{c} [v'^{-1/2}]_{v_0}^v = \frac{2m}{c} \left(\frac{1}{\sqrt{v}} - \frac{1}{\sqrt{v_0}} \right)$$

or, solving for v , $v = v_0/(1 + ct\sqrt{v_0}/2m)^2$. Clearly, $v = 0$ only when $t \rightarrow \infty$.

2.9 * If we integrate the given equation, with the velocity running from v_{y0} to v_y and time from 0 to t , we find $m \ln[(v_y - v_{\text{ter}})/(v_{y0} - v_{\text{ter}})] = -bt$, or, solving for v_y ,

$$v_y = v_{\text{ter}} + (v_{y0} - v_{\text{ter}})e^{-bt/m},$$

which is exactly (2.30), since $m/b = \tau$.

2.10 ** (a) The characteristic time is $\tau = m/b = m/(3\pi\eta D) = 1.4 \times 10^{-4}$ s. The buoyant force on the ball bearing is $F_b = \rho_g Vg$, where ρ_g is the density of glycerin and V is the volume of the ball bearing. This can be rewritten as $F_b = (\rho_g/\rho_s)mg$, where ρ_s is the density of steel and m is the mass of the ball bearing. This upward force must be added (with a minus sign) to the right side of the equation of motion (2.25). Therefore the terminal speed changes from $v_{\text{ter}} = mg/b$ to

$$v_{\text{ter}} = \frac{mg - F_b}{b} = \frac{mg(1 - \rho_g/\rho_s)}{b} = \tau g(1 - \rho_g/\rho_s) = 1.2 \text{ mm/s}.$$

The time to reach 95% of v_{ter} is $3\tau = 4.3 \times 10^{-4}$ s. [As you can easily check, the buoyant force changes the terminal speed but not the exponent $-t/\tau$ in Eq.(2.33).]

$$\mathbf{(b)} \quad \frac{f_{\text{quad}}}{f_{\text{lin}}} = \frac{\kappa \rho_g A v^2}{3\pi\eta D v} = \frac{\frac{1}{4}\rho_g (\frac{1}{4}\pi D^2)v}{3\pi\eta D} = \frac{\rho_g D v}{48\eta} = 5 \times 10^{-6},$$

so it was an excellent approximation to neglect f_{quad} .

2.14 *** (a) With $F = -F_0 e^{v/V}$, the second law ($mdv/dt = F$) separates to become $me^{-v/V} dv = -F_0 dt$. This is easily integrated (from time 0 to t or v_0 to v) and the result solved to give

$$v = -V \ln \left(\frac{F_0 t}{mV} + e^{-v_0/V} \right).$$

(b) This is zero when the argument of the log is 1, so that $t = (1 - e^{-v_0/V})mV/F_0$.

(c) Using the result that $\int \ln(x) dx = x \ln(x) - x$, we can evaluate the integral $x = \int v dt$ to give

$$x(t) = Vt - \frac{mV^2}{F_0} \left[\left(\frac{F_0 t}{mV} + e^{-v_0/V} \right) \ln \left(\frac{F_0 t}{mV} + e^{-v_0/V} \right) + \frac{v_0}{V} e^{-v_0/V} \right]$$

and, substituting the time from part (b),

$$x_{\max} = \frac{mV^2}{F_0} \left[1 - e^{-v_0/V} \left(1 + \frac{v_0}{V} \right) \right].$$

2.15 * Since the only force is the projectile's weight mg , Newton's second law implies that $\ddot{\mathbf{r}} = \mathbf{g}$ and its two components can be integrated twice to give the well-known results $x = v_{x0}t$ and $y = v_{y0}t - \frac{1}{2}gt^2$ (if we take $x_0 = y_0 = 0$). At landing, $y = 0$, which gives the time of flight as $t = 2v_{y0}/g$. The range is just the value of x at this time, namely $x = v_{x0}t = 2v_{x0}v_{y0}/g$.

2.16 * As usual, $x = (v_0 \cos \theta)t$ and $y = (v_0 \sin \theta)t - \frac{1}{2}gt^2$. The time to reach the plane of the wall ($x = d$) is $t = d/(v_0 \cos \theta)$ and the ball's height at that time is $y = d \tan \theta - \frac{1}{2}gd^2/(v_0 \cos \theta)^2$. Notice that this height decreases monotonically as v_0 decreases. Thus there is indeed a minimum speed $v_0(\min)$ for which the ball clears the wall. Putting $y = h$ and solving for v_0 we find that

$$v_0(\min) = \sqrt{\frac{gd^2}{2(d \tan \theta - h) \cos^2 \theta}}.$$

If $\tan \theta < h/d$, the argument of the square root is negative and there is no real $v_0(\min)$; physically, the ball's initial velocity is aimed below the top of the wall, so the ball cannot possibly clear the wall whatever its speed. With the given numbers, $v_0(\min) = 26.4$ m/s or roughly 50 mi/hr.

2.17 * From the first of Equations (2.36) we find that $1 - e^{-t/\tau} = x(t)/(v_{x0}\tau)$ and hence $t = -\tau \ln(1 - x/v_{x0}\tau)$. Substituting these into the second of (2.36), we find

$$y = \frac{v_{y0} + v_{\text{ter}}}{v_{x0}} x + v_{\text{ter}} \tau \ln \left(1 - \frac{x}{v_{x0}\tau} \right)$$

which is (2.37).